Customer Behaviour & Revenue Insights Analysis

A Data-Driven Report Based on Python, SQL, and Power BI

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This report presents a comprehensive analysis of customer shopping behaviour using Python-based exploratory data analysis (EDA), SQL-driven business queries, and a fully interactive Power BI dashboard. The objective is to uncover insights into revenue drivers, customer segments, subscription impact, discount dependency, and product/category performance to support strategic data-driven decisions.

Tools Used: Python (Pandas), SQL (MySQL), Power BI, Overleaf (Reporting)
Techniques Applied: EDA, Feature Engineering, Segmentation, KPI Analysis, Dashboarding

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1 Executive Summary

Executive Summary This project analyzes transactional customer data to uncover key patterns in spending behavior, subscription impact, discount usage, seasonal demand, and product/category performance. The goal is to translate raw data into insights that support strategic decision-making across marketing, inventory planning, and customer retention.

Key highlights:

- Total customers : 15000
- Core takeaways :
  - Average Purchase Amount: $ 56.97
  - Average Review Rating overall: 3.74
  - Top Category by revenue: Clothing
  - Highest-Demand Season: Winter

Recommendation at a glance: Boost subscription adoption, refine discount policies to protect margins, and align inventory planning with strong seasonal demand patterns.

2 Project Overview

2.1 Objectives

1. Identify which customers and categories drive revenue and how subscription status affects spend and loyalty.
2. Measure discount dependency and its impact on average order value.
3. Build a stakeholder-ready Power BI dashboard and produce actionable recommendations.
4. Deliver reproducible SQL & EDA artifacts.

2.2 Scope

The analysis uses transactional records with the following feature groups: demographics (age, gender, location), product attributes, transaction metrics (`purchase_amount`), behavior flags (discount, subscription), and customer history (`previous_purchases`). More details and initial EDA steps are in the project PDF.

3 Dataset Summary

3.1 High-level metadata

- **Rows: 1500**
- **Columns:** 18
- **Notable fields:** `customer_id`, `age`, `gender`, `item_purchased`, `category`, `purchase_amount`, `subscription_status`, `discount_applied`, `review_rating`, `previous_purchases`, `purchase_frequency_days`, `shipping_type`, `location`.

3.2 Missing data & cleaning highlights

- **Review rating** had missing values — imputed using median rating per product category (see `python_eda.ipynb`).
- Dropped `promo_code_used` after verifying redundancy with `discount_applied`.
- Engineered `age_group` and `purchase_frequency_days`.
- Standardized categorical labels (e.g. shipping types, size labels).

3.3 Data dictionary (short)

Column	**Description**
customer_id	Unique customer identifier
purchase_amount	Transaction value (currency)
category	Product category (Clothing, Accessories, Footwear, Outerwear)
subscription_status	'Yes' / 'No' flag for subscription
discount_applied	'Yes' / 'No' flag for discount use
review_rating	Numeric rating provided by customers
previous_purchases	Prior purchases by customer (integer)
purchase_frequency_days	Days between purchases (derived)

season	Season of purchase (Spring, Summer, Fall, Winter)
shipping_type	Shipping method (Standard, Express, etc.)
...	...

4 Data Preparation & EDA (Summary)

This section summarizes the EDA steps performed in Python (see `python_eda.ipynb` for the code).

4.1 Key transformations

- Handled nulls in `review_rating` by per-category median imputation.
- Built `age_group` bins: Young Adult, Adult, Middle-aged, Senior.

	age	age_group
0	43	Adult
1	63	Senior
2	30	Young Adult
3	24	Young Adult
4	33	Adult
5	73	Senior
6	54	Middle-aged
7	37	Adult
8	22	Young Adult
9	25	Young Adult

Figure 1: Checking Null values

- Created `purchase_frequency_days` from `frequency_of_purchases`.

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	365	Annually
2	90	Quarterly
3	365	Annually
4	7	Weekly
5	14	Fortnightly
6	90	Quarterly
7	90	Quarterly
8	7	Weekly
9	90	Every 3 Months

Figure 2: Checking Null values

5 SQL-Based Business Findings

5.1 Revenue by demographics

Question: How does total revenue vary by gender and age group?

Gender	Revenue
Male	584313
Female	270235

Figure 3: Revenue by Gender

5.2 Subscription impact

Question: Do subscribed customers spend more? (Avg spend and total revenue)

Customer_Status	Total_Customers	Average_Spend	Total_Revenue
Non-Subscribers	8335	57.40	478458
Subscribers	6665	56.43	376090

Figure 4: Subscribers vs Non-Subscribers

5.3 Discount behaviour

Question: Which product categories depend heavily on discounts?

Category	Discount_Dependency_Ratio
Accessories	0.4794
Footwear	0.4792
Clothing	0.4787
Outerwear	0.4645

Figure 5: Discount Dependency by Category

5.4 Top-rated and top-selling products

Question: Which products have the highest average ratings and which sell the most?

Product	Avg_Product_Rating
Skirt	3.79
Jewelry	3.78
Handbag	3.78
Coat	3.77
Socks	3.77

Figure 6: Top 5 products by average rating

Products	Volume
Pants	734
Shirt	721
Hoodie	718
Sandals	702
Backpack	694
T-shirt	684
Belt	676
Sweater	647
Socks	647
Shorts	640

Figure 7: Top 10 best-selling products:

5.5 Customer Segmentation

Question: New / Returning / Loyal segmentation and their revenue contribution.

Customer_Segment	Number_of_Customers	Total_Revenue
Loyal	13249	752658
Returning	1454	84984
New	297	16906

Figure 8: Customer Segments: New / Returning / Loyal (placeholder)

Question: What is the revenue contribution and Average Order Value (AOV) of each age group ?

Age_Group	Total_Revenue	Average_Order_Value
Adult	223401	57.03
Young Adult	215488	57.40
Middle-aged	211502	57.02
Senior	204157	56.40

Figure 9: Customer Segments: New / Returning / Loyal (placeholder)

5.6 Seasonal Impact

Question: Which seasons generate the highest revenue for each category?

Category	Season	Total_Revenue
Clothing	Winter	104546
Accessories	Summer	56130
Footwear	Fall	38557
Outerwear	Spring	24536

Figure 10: Highest Revenue For Each Category

6 Dashboard (Power BI)

- **Top KPIs:** total customers, avg purchase amount, avg review rating.
- **Filters:** subscription status, gender, category, shipping type.

- **Charts:** revenue by category, sales by category, revenue by age group, sales by age group.

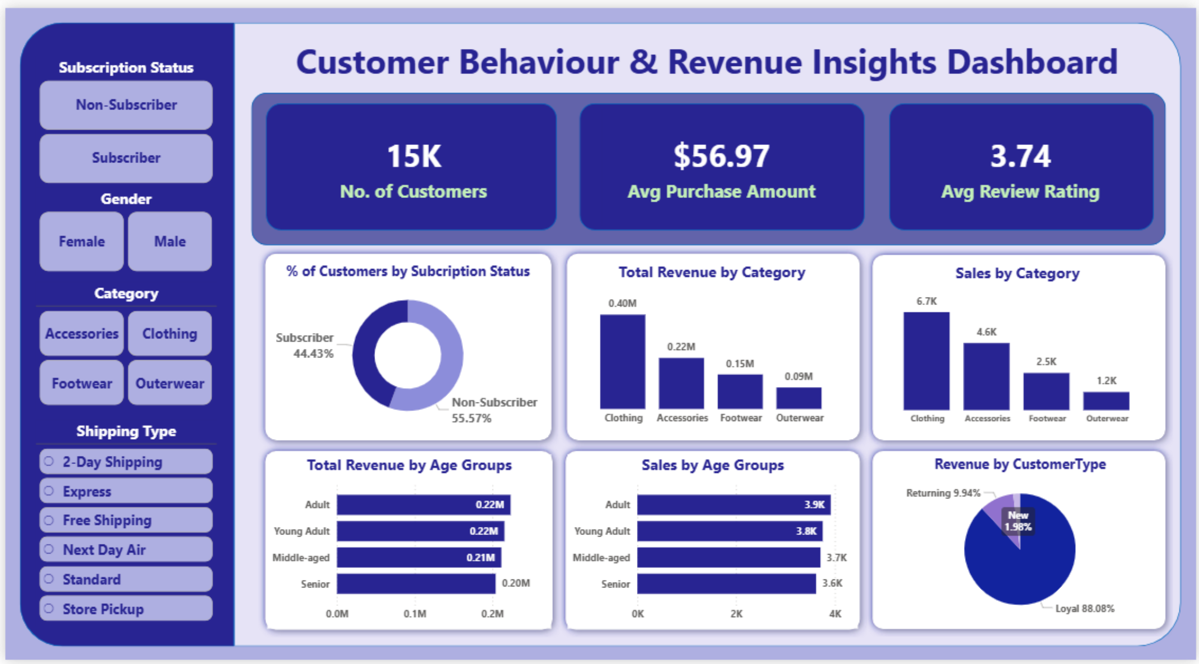

Figure 11: Power BI dashboard

7 Key Insights (Actionable)

- **Subscription impact:** Subscribers (avg spend = \$ 478458) contribute 55.98% of revenue.
- **Top category: Clothing** drives **46.85%** of total revenue.
- **Discount dependency:** Category **Accessories** has a discount dependency ratio of **0.4794**.
- **Seasonality:** Highest-revenue season for category **Clothing** is **Winter**.
- **Customer loyalty:** Loyal customers (`previous_purchases` $\geq$ 10) account for **88.08%** % of revenue.

8 Business Recommendations

Prioritized recommendations based on the analysis:

1. **Boost Subscriptions:** Offer trial discounts or time-limited perks; prioritize channels where subscribers are more concentrated.

2. **Refine Discount Strategy:** Reduce blanket discounts on categories with high discount dependency and low margin; A/B test targeted discounts on price-sensitive segments.

3. **Inventory & Seasonality:** Increase stock of top-performing seasonal items prior to peak season and reduce SKUs with consistently low sales.

4. **Customer Loyalty Program:** Create micro-incentives that move Returning customers to Loyal (e.g., small reward after 3 purchases).

5. **Shipping Optimization:** If Express correlates with higher spend, explore upsell messaging for faster shipping.